A3144 OH3144   Y3144 Hall sensor TO-92UA (10PCS/LOT)

Product Photos	TO-92 Pkg
RoHS Information	HF/Green Compliance Declaration
Catalog Drawings	UA-92 Series Side
Category	Sensors, Transducers
Family	Magnetic Sensors - Hall Effect, Digital Switch, Linear, Compass (ICs)
Series	-
Packaging	Bulk
Sensing Range	9.5mT Trip, -9.5mT Release
Type	Bipolar Latch
Voltage - Supply	3.5 V ~ 24 V
Current - Supply	5mA
Current - Output (Max)	50mA
Output Type	Digital, Open Drain
Features	-
Operating Temperature	-40°C ~ 150°C
Package / Case	TO-226-3, TO-92-3 Short Body
Supplier Device Package	TO-92UA

49E is a small, multi-function linear Hall, its input is the magnetic induction intensity, and the output is a voltage proportional to the input. The integrated circuit has a low noise output, which makes it unnecessary to use external filtering. It also includes precision resistors for more temperature stability and accuracy.

Features

Single current source output
low noise output
Both positive and negative magnetic fields can be sensed

typical application
Magnetic reader
Current Detection
motor control
position detection
Liquid level sensing
weight sensing
Vibration sensor
ferrous metal detector
Electric bike governor
Other applications for detecting magnetic fields.
Limit parameters

voltage V_{CC} 10.V
range of working temperature T_A .... -20～100℃
Storage temperature T_S .... -50～150℃

(T_A=25℃,Vcc=5.0V)

Parameter	Symbol	Test Conditions	Minimum	Typical value	Maximum value	Unit
Operating Voltage	Vcc		2.5	5	10	V
Supply current	Icc		-	4.2	8	mA
Quiescent output voltage	VNULL	@ B=0GS	2.35	2.5	2.65	V
Output Voltage Sensitivity	S	B=±100GS	1.8	2.0	2.2	mV/GS
Output high level	VH	B=+1200GS	-	-	4.2	V
Output low level	VL	B=-1200GS	0.8	-	-	V
Output resistance	RO			40	100	Ω
Magnetic field range	B		-	±1200		GS
Output noise	BW=10Hz to 10kHz			90		μV

Overview：

41 Hall lock IC is composed of reverse voltage protector, voltage regulator, Hall voltage generator, differential amplifier, Schmitt trigger and output stage, which can convert the changing magnetic field signal into digital

voltage output.

Features：

High sensitivity, anti-stress, wide voltage range, guaranteed locking under high temperature and high pressure

typical application：

Highly sensitive non-contact switch, DC brushless motor, DC brushless fan limit parameters (25 ℃)

Voltage	VCC	4.5~24V
Output current	IO	25mA
Working temperature	TA	-40~150℃
Storage temperature range	TS	-55~150℃

General Description US1881 The US1881 is a bipolar Hall effect sensor IC fabricated from mixed signal CMOS technology. It incorporates advanced chopper stabilization techniques to provide accurate and stable magnetic switch points. There are many applications for this HED in addition to those listed above. The design, specifications and performance have been optimized for commutation applications in 5V and 12V brushless DC motors.The output transistor will be latched on (BOP) in the presence of a sufficiently strong South pole magnetic field facing the marked side of the package. Similarly, the output will be latched off (BRP) in the presence of a North field. The SOT-23 device is reversed from the UA package.The SOT-23 output transistor will be latched on in the presence of a sufficiently strong North pole magnetic field applied to the marked face.

Applications

-Solid state switch

-Brushless DC motor commutation
-Speed sensing
-Linear position sensing
-Angular position sensing
-Current sensing

Features-Chopper stabilized amplifier stage-Optimized for BDC motor applications-New miniature package / thin, high reliability package-Operation down to 3.5V-CMOS for optimum stability, quality and cost**Functional Block DiagramsAbsolute Maximum Ratings (at Ta=25℃)**

Characteristics	Symbol	Values	Unit
Supply Voltage	Vdd	3.5~24	V
Supply Current	IDD	50	mA
Output Voltage	VOUT	3.5~24	V
Output Current	IOUT	50	mA
Power Dissipation	PD	100	mW
Operating Temperature Range	Ta	-40~+150	°C

Overview:

Product model: OH137A0 Working temperature: -40~85°C, package TO-92S

The OH137A0 unipolar Hall switch circuit is composed of reverse voltage protector, voltage regulator, Hall voltage generator, differential amplifier, Schmitt trigger and open collector output stage, which can convert the changing magnetic field signal into digital voltage output. Because of its reliable and stable performance, it is widely used in various fields as a non-contact switch.

Features:

Fast switching speed, no instant jitter

Wide operating frequency and good consistency

Bipolar non-locking

The circuit can interface directly with various logic circuits

Achievable functions:

Contactless switch

position detection

Speed detection

Traffic detection

Voltage	VCC	4.5～24V
Output current	IO	25mA
Working temperature	TA	-40～85℃
Storage temperature range	TS	-55～150℃